

Project Charter

Project Purpose

The purpose of this project is to design, implement and validate a secure enterprise network infrastructure for a smart manufacturing facility. The project simulates a real-world IT environment with an emphasis on security, segmentation, and audit ready documentation.

Project Objectives & MVP

This project's objectives, which must all be completed to meet minimum viable product requirements, are as follows:

- Translate OT/IT Segregation (NIST CSF-lite) requirements into implementable controls and measurable evidence
- Design segmented Layer-2/Layer-3 architecture with a clean IP plan and verification tests
- Deploy Windows Server services (AD/DNS/DHCP) and security-focused GPO baselines
- Implement firewall/NAT/ACL policy using least privileged and documented test cases
- Create a small Python automation deliverable with logged, repeatable output
- Produce runbooks, diagrams, and a final report suitable for an audit-style review

Scope

In Scope:

- Runbooks, rebuild guide, risk register, change log
- Python automation proof of concept
- VLANs, Trunking, Inter-VLAN routing
- OT/IT segregation (NIST CSF-lite) control mapping and evidence
- Window services (AD, DNS, DHCP, baseline GPOs)

Out of scope:

- Live production deployment

- Custom application development

Success Criteria

- All required deliverables meet rubric and deadline requirements
- Environment can be rebuilt using documented procedures
- Security controls are mapped to requirements with supporting evidence
- Routing, identity services (AD/DNS/DHCP) and firewall controls pass documented test cases

Deliverables

- Project charter
- IP plan and VLAN list
- Logical/physical network diagrams
- Packet tracer models and VM configuration
- Runbooks, rebuild guide, final report, presentation, evidence bundle
- Python automation script with sample output

Communication Plan

- Tools: teams/discord
- Meeting: meet on campus - 3 hours per week/ outside class – 1 hour
- Reporting: evidence documented in repo every session
- Instructor: check in with instructor for weekly progress and milestone reviews